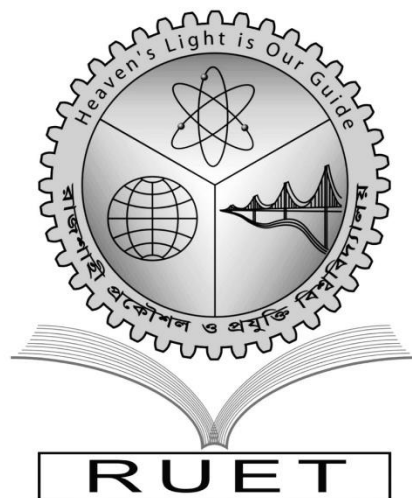


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**Rajshahi University of Engineering & Technology**

**Department of Computer Science & Engineering**



**Sessional (Lab Report)-02**

**Course Name** : Database Systems Sessional  
**Course Code** : CSE 3102  
**Experiment No** : 02  
**Experiment Name** : Self Query generator using a database  
**Date of Experiment** : 16 Jun, 2025  
**Date of Submission** : 18 Jun, 2025

|                                                                                                     |                                                                                                                                          |
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**1. Create a database called “CricketerStatsDB”.**

| SQL Command                                              |
|----------------------------------------------------------|
| Create Schema CricketerStatsDB;<br>use cricketerstatsdb; |

**2. Create a table titled “Players” with the following constraints.**

| Field         | Type          | Constraints                           |
|---------------|---------------|---------------------------------------|
| player_id     | INT           | PRIMARY KEY, NOT NULL, AUTO_INCREMENT |
| player_name   | VARCHAR (100) | NOT NULL                              |
| country       | VARCHAR (50)  | NOT NULL                              |
| date_of_birth | DATE          | NOT NULL                              |
| batting_style | VARCHAR (20)  | NOT NULL                              |
| bowling_style | VARCHAR (50)  | NULLABLE                              |

| SQL Command                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| create table Players(<br>player_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,<br>player_name VARCHAR (100) NOT NULL,<br>country VARCHAR (50) NOT NULL,<br>date_of_birth DATE NOT NULL,<br>batting_style VARCHAR (20) NOT NULL,<br>bowling_style VARCHAR (50) NULL<br>); |

**3. Create a table titled “Matches” with the following constraints.**

| Field      | Type          | Constraints                           |
|------------|---------------|---------------------------------------|
| match_id   | INT           | PRIMARY KEY, NOT NULL, AUTO_INCREMENT |
| match_date | DATE          | NOT NULL                              |
| venue      | VARCHAR (100) | NOT NULL                              |
| match_type | VARCHAR (20)  | NOT NULL                              |

| SQL Command                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| create table Matches(<br>match_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,<br>match_date DATE NOT NULL,<br>venue VARCHAR (100) NOT NULL,<br>match_type VARCHAR(20) NOT NULL<br>); |

4. Create a table titled “Performances” with the following constraints.

| Field          | Type          | Constraints                           |
|----------------|---------------|---------------------------------------|
| performance_id | INT           | PRIMARY KEY, NOT NULL, AUTO_INCREMENT |
| player_id      | INT           | NOT NULL                              |
| match_id       | INT           | NOT NULL                              |
| runs_scored    | INT           | NOT NULL DEFAULT 0                    |
| balls_faced    | INT           | NOT NULL DEFAULT 0                    |
| wickets_taken  | INT           | NOT NULL DEFAULT 0                    |
| overs_bowled   | DECIMAL (4,1) | NOT NULL DEFAULT 0.0                  |

| SQL Command                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>create table Performances(<br/>  performance_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,<br/>  player_id INT NOT NULL,<br/>  match_id INT NOT NULL,<br/>  runs_scored INT NOT NULL DEFAULT 0,<br/>  balls_faced INT NOT NULL DEFAULT 0,<br/>  wickets_taken INT NOT NULL DEFAULT 0,<br/>  overs_bowled DECIMAL (4,1) NOT NULL DEFAULT 0<br/>);</pre> |

5. Populate the table “Players” with the following data

| player_name     | country     | date_of_birth | batting_style | bowling_style |
|-----------------|-------------|---------------|---------------|---------------|
| Virat Kohli     | India       | 1988-11-05    | Right-Handed  | NULL          |
| Joe Root        | England     | 1990-12-30    | Right-Handed  | Off Spin      |
| David Warner    | Australia   | 1986-10-27    | Left-Handed   | NULL          |
| Shaheen Afridi  | Pakistan    | 2000-04-06    | Left-Handed   | Left-arm Fast |
| Kane Williamson | New Zealand | 1990-08-08    | Right-Handed  | Off Spin      |

| SQL Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>INSERT INTO Players (player_name, country, date_of_birth, batting_style,<br/>bowling_style)<br/>VALUES<br/>( 'Virat Kohli', 'India', '1988-11-05', 'Right-Handed', NULL),<br/>( 'Joe Root', 'England', '1990-12-30', 'Right-Handed', 'Off Spin'),<br/>( 'David Warner', 'Australia', '1986-10-27', 'Left-Handed', NULL),<br/>( 'Shaheen Afridi', 'Pakistan', '2000-04-06', 'Left-Handed', 'Left-arm Fast'),<br/>( 'Kane Williamson', 'New Zealand', '1990-08-08', 'Right-Handed', 'Off Spin');</pre> |

6. Populate the table “Matches” with the following data.

| match_date | venue                     | match_type |
|------------|---------------------------|------------|
| 2024-03-15 | MCG, Australia            | Test       |
| 2024-04-02 | Lord's, England           | ODI        |
| 2024-04-18 | Eden Gardens, India       | T20        |
| 2024-05-01 | Gaddafi Stadium, Pakistan | Test       |

| SQL Command                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INSERT INTO Matches (match_date, venue, match_type)<br>VALUES<br>( '2024-03-15', 'MCG, Australia', 'Test'),<br>( '2024-04-02', 'Lord"s, England', 'ODI'),<br>( '2024-04-18', 'Eden Gardens, India', 'T20'),<br>( '2024-05-01', 'Gaddafi Stadium, Pakistan', 'Test'); |

7. Populate the table “Performances” with the following data.

| player_id | match_id | runs_scored | balls_faced | wickets_taken | overs_bowled |
|-----------|----------|-------------|-------------|---------------|--------------|
| 1         | 1        | 120         | 200         | 0             | 0.0          |
| 2         | 1        | 150         | 250         | 0             | 5.0          |
| 3         | 2        | 80          | 70          | 0             | 0.0          |
| 4         | 4        | 15          | 10          | 5             | 20.0         |
| 5         | 3        | 45          | 30          | 0             | 2.0          |
| 1         | 3        | 60          | 45          | 0             | 0.0          |
| 2         | 4        | 55          | 50          | 0             | 4.0          |

| SQL Command                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INSERT INTO Performances (player_id, match_id, runs_scored, balls_faced,<br>wickets_taken, overs_bowled)<br>VALUES<br>(1, 1, 120, 200, 0, 0.0),<br>(2, 1, 150, 250, 0, 5.0),<br>(3, 2, 80, 70, 0, 0.0),<br>(4, 4, 15, 10, 5, 20.0),<br>(5, 3, 45, 30, 0, 2.0),<br>(1, 3, 60, 45, 0, 0.0),<br>(2, 4, 55, 50, 0, 4.0); |

## Newly Generated Queries & Output Based on “CricketerStatsDB” Database

**Query-1:** Show the tuple/tuples of the players whose batting style is right-handed.

```
select *  
from players  
where batting_style='right-handed'
```

Output:

| player_id | player_name     | country     | date_of_birth | batting_style | bowling_style |
|-----------|-----------------|-------------|---------------|---------------|---------------|
| 1         | Virat Kohli     | India       | 1988-11-05    | Right-Handed  | NULL          |
| 2         | Joe Root        | England     | 1990-12-30    | Right-Handed  | Off Spin      |
| 5         | Kane Williamson | New Zealand | 1990-08-08    | Right-Handed  | Off Spin      |
| NULL      | NULL            | NULL        | NULL          | NULL          | NULL          |

**Query-2:** Show the tuple/tuples of the players along with player’s age with the format like “X years Y months”. (Here X and Y are positive integer numbers.)

```
select *,concat(year(current_date)-year(date_of_birth)-1,' years ', month(current_date)-  
month(date_of_birth)+12,' months ') as age  
from players  
where month(current_date)-month(date_of_birth)<0  
union  
select *,concat(year(current_date)-year(date_of_birth),' years ', month(current_date)-  
month(date_of_birth),' months ') as age  
from players  
where month(current_date)-month(date_of_birth)>=0
```

Output:

| player_id | player_name     | country     | date_of_birth | batting_style | bowling_style | age                |
|-----------|-----------------|-------------|---------------|---------------|---------------|--------------------|
| 1         | Virat Kohli     | India       | 1988-11-05    | Right-Handed  | NULL          | 36 years 7 months  |
| 2         | Joe Root        | England     | 1990-12-30    | Right-Handed  | Off Spin      | 34 years 6 months  |
| 3         | David Warner    | Australia   | 1986-10-27    | Left-Handed   | NULL          | 38 years 8 months  |
| 5         | Kane Williamson | New Zealand | 1990-08-08    | Right-Handed  | Off Spin      | 34 years 10 months |
| 4         | Shaheen Afridi  | Pakistan    | 2000-04-06    | Left-Handed   | Left-arm Fast | 25 years 2 months  |

**Query-3:** Show the player id, player name, total runs, total ball faced and strike rate in a table from maximum to minimum

```
select player_id,player_name,sum(runs_scored) as Runs,sum(balls_faced) as Balls,  
sum(runs_scored)/sum(balls_faced)*100 as Strike_Rate  
from players natural join performances  
group by player_id  
order by Strike_Rate desc
```

Output:

| player_id | player_name     | Runs | Balls | Strike_Rate |
|-----------|-----------------|------|-------|-------------|
| 4         | Shaheen Afridi  | 15   | 10    | 150.0000    |
| 5         | Kane Williamson | 45   | 30    | 150.0000    |
| 3         | David Warner    | 80   | 70    | 114.2857    |
| 1         | Virat Kohli     | 180  | 245   | 73.4694     |
| 2         | Joe Root        | 205  | 300   | 68.3333     |

**Query-4:** find the total runs scored in test cricket format.

```
select match_type, sum(runs_scored) as Total_runs  
from matches natural join performances  
group by match_type  
having match_type='test'
```

Output:

| match_type | Total_runs |
|------------|------------|
| Test       | 340        |

**Query-5:** Find the id, name of the player who has scored maximum runs in T20 in a single match.

```
select player_id,player_name,runs_scored  
from matches natural join performances natural join players  
where runs_scored=  
(select max(runs_scored)  
from matches natural join performances  
where match_type='T20'  
)
```

Output:

| player_id | player_name | runs_scored |
|-----------|-------------|-------------|
| 1         | Virat Kohli | 60          |

**Query-6: Show all the data of the last match**

```
select *
from matches natural join performances
where match_id=
(select match_id
from matches
order by match_date desc
limit 1)
```

Output:

| match_id | match_date | venue                     | match_type | performance_id | player_id | runs_scored | balls_faced | wickets_taken | overs_bowled |
|----------|------------|---------------------------|------------|----------------|-----------|-------------|-------------|---------------|--------------|
| 4        | 2024-05-01 | Gaddafi Stadium, Pakistan | Test       | 4              | 4         | 15          | 10          | 5             | 20.0         |
| 4        | 2024-05-01 | Gaddafi Stadium, Pakistan | Test       | 7              | 2         | 55          | 50          | 0             | 4.0          |

**Query-7: Show all the data where the venue country is Australia or Pakistan**

```
select *
from matches natural join performances natural join players
where venue like ('%Australia%') or venue like ('%Pakistan%')
```

Output:

| player_id | match_id | match_date | venue                     | match_type | performance_id | runs_scored | balls_faced | wickets_taken |
|-----------|----------|------------|---------------------------|------------|----------------|-------------|-------------|---------------|
| 1         | 1        | 2024-03-15 | MCG, Australia            | Test       | 1              | 120         | 200         | 0             |
| 2         | 1        | 2024-03-15 | MCG, Australia            | Test       | 2              | 150         | 250         | 0             |
| 4         | 4        | 2024-05-01 | Gaddafi Stadium, Pakistan | Test       | 4              | 15          | 10          | 5             |
| 2         | 4        | 2024-05-01 | Gaddafi Stadium, Pakistan | Test       | 7              | 55          | 50          | 0             |

| overs_bowled | player_name    | country  | date_of_birth | batting_style | bowling_style |
|--------------|----------------|----------|---------------|---------------|---------------|
| 0.0          | Virat Kohli    | India    | 1988-11-05    | Right-Handed  | NULL          |
| 5.0          | Joe Root       | England  | 1990-12-30    | Right-Handed  | Off Spin      |
| 20.0         | Shaheen Afridi | Pakistan | 2000-04-06    | Left-Handed   | Left-arm Fast |
| 4.0          | Joe Root       | England  | 1990-12-30    | Right-Handed  | Off Spin      |